

HOMEWORK 1

System Setup: Install Ubuntu / FreeBSD

ytshih, hyguo

國立陽明交通大學資工系資訊中心
Information Technology Center, Department of Computer Science, NYCU

Outline

- System installation
 - Install FreeBSD 14.1-RELEASE (alternatively Ubuntu 24.04 LTS)
- Basic setup
 - Set hostname
 - Create user and group
- Motd
- Change mirror site
- Setup WireGuard®
- Rules
- Appendix: How to use NASA Online Judge

Basic Setup (1/2)

- Your machine should boot using UEFI (5%)
- Set hostname to `sa2024-${ID}` (5%)
- Create user
 - Create a **user named judge for** online judge
 - Should be a member of `wheel` group (or `sudo` on Ubuntu)
 - Should be able to **execute sudo** commands **without entering password** (15%)
 - Should use **sh** as the **default shell** (10%)
 - You should use this user instead of root for subsequent operations
- Create group
 - Create a group `nycusa` (5%)
 - Add `judge` user to this group (5%)

Basic Setup (2/2)

- Set your machine to Taiwan Standard Time (10%)
 - Sync time with servers from National Standard Time and Frequency Laboratory (國家時間與頻率標準實驗室)
- Secure Shell (15%)
 - Enable sshd
 - Install this public key to your /home/judge/.ssh for Online Judge

```
$ wget https://nasa.cs.nycu.edu.tw/sa/2024/nasakey.pub  
$ cat nasakey.pub >> /home/judge/.ssh/authorized_keys
```

- Verify the fingerprint of public key

```
$ ssh-keygen -l -f nasakey.pub  
256 SHA256:l2xVg+C+hMjMldX6htc4SUPE5taFsxKkevTgiGmpeHA judge@sa-2024  
(ED25519)
```

Motd

- Abbreviation of “message of the day”
- Shown upon login every time
- Example (Ubuntu)

```
$ ssh judge@localhost
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-41-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

... (The rest of output is truncated for simplicity)
```

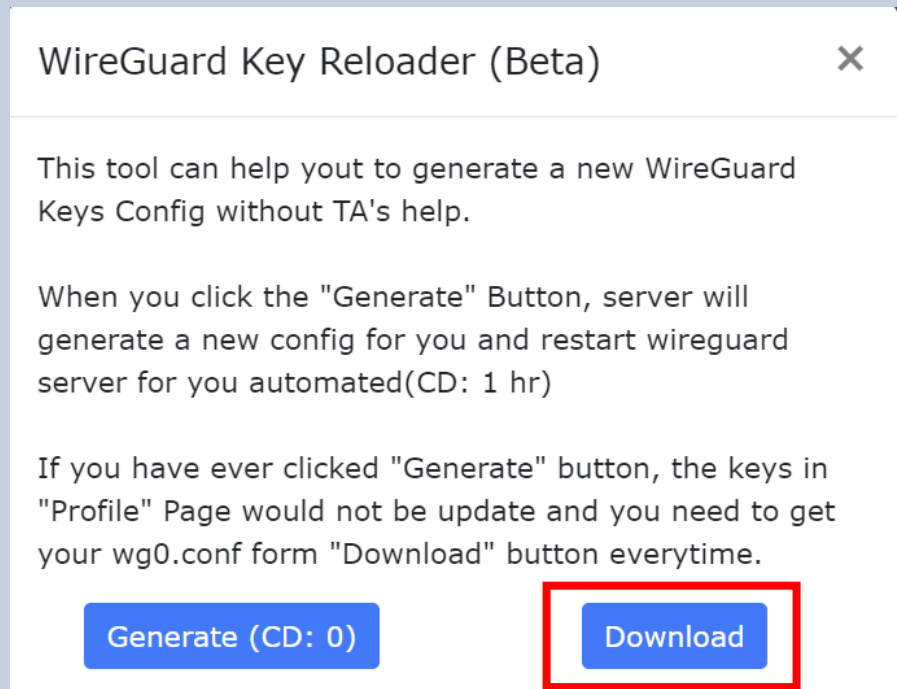
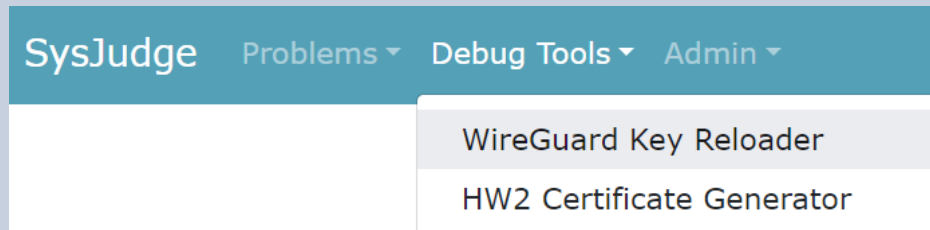
- Edit motd to show “NYCU-SA-2024-`${Student ID}`” (5%)
 - Hint: consult the manual [motd\(5\)](#)

Package Repository Mirror

- As a system administrator, you are responsible for managing packages on a system
- CSIT provides mirrors for a variety of Linux distributions, serving boot images, packages, and even Linux Kernel sources!
 - <https://it.cs.nycu.edu.tw/equipment-linux-mirror>
- Generally, choosing mirrors nearby improves download speed
- Configure your package manager to use CSIT mirror (10%)
 - For FreeBSD systems, use one of the mirror servers in Japan
 - <http://pkg0.kwc.freebsd.org/>
 - <http://pkg0.bbt.freebsd.org/>

Setup WireGuard® (1/3)

- Install WireGuard® on your system
 - Refer to [Installation guide](#)
- Log in to [NASA Online Judge](#) to get your configuration



Example WireGuard® Configuration

wg0.conf

[Interface]

PrivateKey = <PRIVATE KEY>

Address = 10.113.<ID>.11/32

[Peer]

PublicKey = <PUBLIC KEY OF SERVER>

AllowedIPs = 10.113.0.0/16

Endpoint = 140.113.168.131:51XXX

PersistentKeepalive = 15

Setup WireGuard® (2/3)

- Configure the connection
 - Configuration directory: `/usr/local/etc/wireguard/`
 - Place the downloaded configuration file to your system
 - Or configure manually:
 - Address: `10.113.${ID}.11/32`
 - Private Key
 - Server Endpoint: `140.113.168.131:51xxx`
 - Public Key
 - AllowedIPs: `10.113.0.0/16`
- Use `wg-quick` to start/stop the connection
 - For detailed usage, please consult the manuals of [`wg-quick\(8\)`](#) and [`wg\(8\)`](#)

Setup WireGuard® (3/3)

- Test connection
 - `ping -c 3 10.113.${ID}.254`
- Online Judge servers can ping your system (15%)

```
$ ping -c 3 10.113.11.254
PING 10.113.11.254 (10.113.11.254) 56(84) bytes of data.
64 bytes from 10.113.11.254: icmp_seq=1 ttl=51 time=3.95 ms
64 bytes from 10.113.11.254: icmp_seq=2 ttl=51 time=5.50 ms
64 bytes from 10.113.11.254: icmp_seq=3 ttl=51 time=5.12 ms

--- 10.113.11.254 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 3.951/4.855/5.498/0.658 ms
```

Rules

- TAs reserve the rights of final explanations
- **Open from 9/12 (Thu) 19:00**
- **Deadline: 9/26 (Thu) 23:59**
- Late submissions will **NOT** be accepted

Attention

- Your work will be scored by Online Judge system
 - Only the **LAST** submission will be scored
 - Late submission will **NOT** be accepted
- **ALWAYS BACKUP** your system before submission, as we may do malicious actions
- Make sure everything works after reboot

Tips

- Install your system on virtual machines to benefit from flexibility
 - Easy install and backup
- Try to make your VM hardware configuration better
 - Disk controller
 - IDE → SATA, NVMe, ...
 - NIC: paravirtualized net, ...

Appendix: How to use Online Judge

國立陽明交通大學資工系資訊中心
Information Technology Center, Department of Computer Science, NYCU

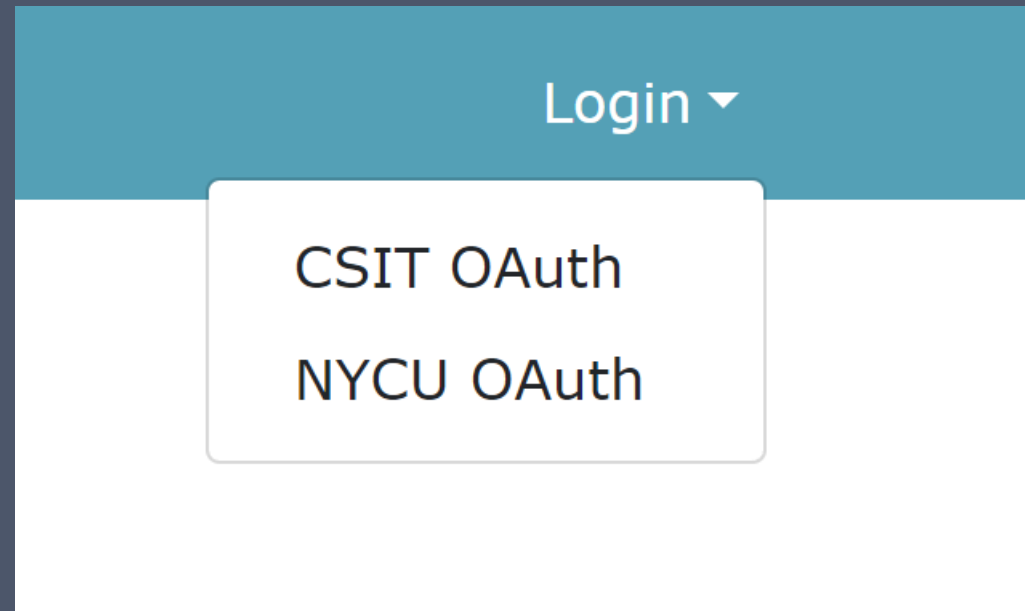
Online Judge

<https://nasaoj.cs.nycu.edu.tw>



Login

Log in NASA Online Judge via either OAuth methods



Troubleshooting WireGuard® Issues

Use key reloader to regenerate configuration to solve VPN-related issues

WireGuard Key Reloader (Beta) ×

This tool can help you to generate a new WireGuard Keys Config without TA's help.

When you click the "Generate" Button, server will generate a new config for you and restart wireguard server for you automated(CD: 1 hr)

If you have ever clicked "Generate" button, the keys in "Profile" Page would not be update and you need to get your wg0.conf from "Download" button everytime.

Generate (CD: 0)

Download

Help

- Join NCTUNASA google group
 - If you have any question, you can post your problem in this group, TAs and Students will help you.
 - <https://groups.google.com/g/nctunasa>
- UNIX 常見指令教學
 - <https://it.cs.nycu.edu.tw/unix-basic-commands>
- How To Ask Questions The Smart Way
 - <https://github.com/ryanhanwu/How-To-Ask-Questions-The-Smart-Way>